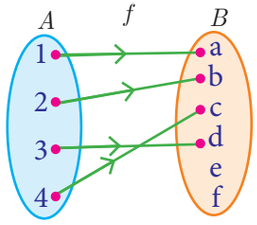
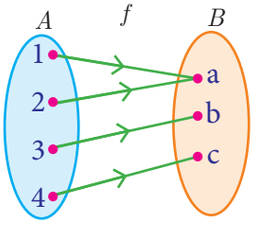
Illustration 15

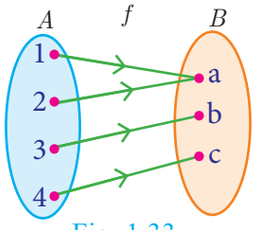
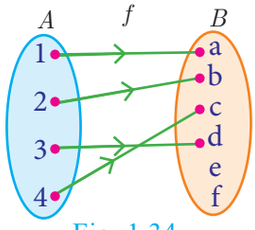
One to One	Many to One
 Fig. 1.31	 Fig. 1.32
Distinct elements of A have distinct images in B.	Two or more elements of A have same image in B.

Note

A one – one and onto function is also called a one – one correspondence.

Thinking Corner

Can there be a one to many function?

Onto	Into
 Fig. 1.33	 Fig. 1.34
Range of f = co-domain (Every element in B has a pre-image in A)	Range of f is a proper subset of co-domain (There exists at least one element in B which is not the image of any element of A)

To determine whether the given function is one–one or not the following test may help us.

1.7.6 Horizontal Line Test

Previously we have seen the vertical line test. Now let us see the horizontal line test. "A function represented in a graph is one–one, if every horizontal line intersects the curve in at most one point".

Example 1.12 Using horizontal line test (Fig.1.35 (a), 1.35 (b), 1.35 (c)), determine which of the following functions are one – one.

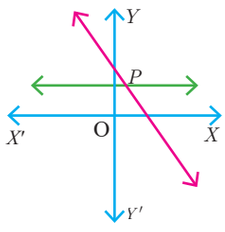


Fig. 1.35(a)

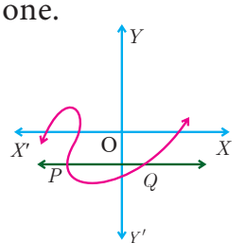


Fig. 1.35(b)

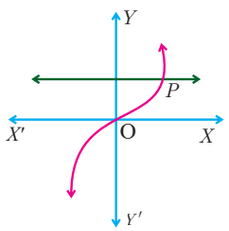


Fig. 1.35(c)

Solution The curves in Fig.1.35 (a) and Fig.1.35 (c) represent a one–one function as the horizontal lines meet the curves in only one point P.

The curve in Fig. 1.35 (b) does not represent a one–one function, since, the horizontal line intersects the curve at two points P and Q.

1 Relation and Functions.indd 20 15-12-2020 02:50:17 PM